ARYAN ZAVERI

Quant Developer | Global Market Modeling | End-to-End Trading Infrastructure
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SUMMARY

Quant Developer with professional algo-trading/quant-dev experience across GreyOak, KIFS and Greeksoft, and a personal multi-asset research/trading infrastructure project. My strongest work is connecting research judgment with systems implementation: quantify relationships across markets, test them, replay them, and wire the surviving logic into risk-aware execution.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Shell, JavaScript
Trading Systems: market data ingestion, live state, feature engineering, signal generation, strategy research, backtesting, walk-forward validation, replay, paper-fill, pre-trade risk, risk managers, order managers, order intents, execution journals, post-trade analytics
Global Modeling: oil and macro themes, Indian/global sector relationships, US/China/Japan/Korea market links, sector rotation, breadth/regime detection, dip-buy/rise-sell behavior, cross-asset correlation, options/futures relationships
Data/Infra: PostgreSQL, Redis, SQLite, JSON contracts, JSONL journals, CSV/TSV archives, Linux, AWS, Git, Docker, WebSockets, REST APIs
Metrics/Analytics: Sharpe, Sortino, profit factor, win rate, drawdown, expectancy, fees, fill rate, margin, deployed capital, capital required, side equity

WORK EXPERIENCE

GreyOak Capital - Quantitative Developer | Mar 2026 - Present

- Build quantitative research systems, market-intelligence infrastructure, market-data services and analyst-facing context-intelligence tooling.
- Work on Horizon, Compass, Context-Intelligence and Ask-GreyOak style systems connecting market context, internal data flows, dashboards, WebSocket/data contracts and research interfaces.
- Own research-to-runtime handoffs across historical archives, live state, signal outputs, runtime logs, validation reports and production-readiness workflows.
- Design data contracts and service handoffs so research outputs can be traced through dashboards, runtime systems and operational reports.

KIFS Trade Capital - Quantitative / Algorithmic Trading Developer | 2024 - Mar 2026

- Built algorithmic trading tools, broker/API workflows, intraday market-data processing, conversion/reversion research utilities and execution analytics.
- Developed intraday/options analytics around tick/depth context, price action, option-chain features, IV/OI/Greeks-style inputs, signal filters and post-market validation.
- Worked on order lifecycle mechanics, execution constraints, latency-aware design and reusable net-PnL evaluation modules for trade-quality analysis.
- Analyzed strategy behavior through charges, win/loss quality, drawdown, expectancy, execution constraints and post-market review workflows.

Greeksoft Technologies - Algorithmic Trading Intern

- Worked around broker-facing algorithmic trading platform workflows: broker connectivity, instrument mapping, market data, order placement, order status and execution flow.
- Gained practical exposure to how broker APIs, authentication, market-data subscriptions and order-state transitions fit inside an algo-trading platform.

PROJECT EXPERIENCE - VAJRA PERSONAL RESEARCH/TRADING INFRASTRUCTURE

Market framework

- Built Vajra around the idea that markets do not move in isolation: Indian sectors, global indices, US/China/Japan/Korea market context, crude oil, FX, macro variables, options/futures structure and crypto risk appetite all feed the research process.
- Focused on quantifying relationships instead of relying on narrative: correlations, lead/lag behavior, breadth, sector themes, volatility regimes, option-chain structure, cross-asset stress, capital usage, execution quality and replay feedback.
- Designed models to compare market state across live execution, intraday signal selection, swing/regime context and capital review, then translate the best available view into risk-aware action.
- Researched themes including dip-buy/rise-sell behavior, momentum, reversal, mean reversion, volatility/options, breadth, regime detection, sector rotation and cross-asset confirmation.

Research-to-execution architecture

- Built the full loop from market data and feature engineering to cross-market relationship analysis, signal ranking, risk decisioning, order intents, execution journals, replay/PnL analytics and model refinement.
- Use Python for research, backtesting, walk-forward studies, orchestration, replay and analytics; use C++ for live socket, risk-manager, order-manager and execution-runtime components.
- Use JSON contracts for signals, risk decisions, order intents, position snapshots, execution events and journals; Redis for hot queues/streams/hashes/positions/heartbeats; PostgreSQL/SQLite for durable candles, instruments, Greeks, snapshots and reports.
- Built clear process boundaries between research, simulation, replay, paper/live-runtime records and post-trade analytics so each result can be interpreted in the right context.

Risk and execution systems

- Built risk around a single independent authority: strategies propose, risk decides, execution implements. Controls cover global capital, segment risk, side exposure, symbol exposure, active-order limits, quote freshness, drawdown state and hard exits.
- Worked on adaptive execution logic: spread/slippage gates, stale-quote rejection, price leaning around bid/ask/mid, replace/adjust loops, chunking, laddering, take-profit, stop-loss, trailing exits and time/hard exits.
- Designed live systems around low-latency state discipline: memory-resident working orders, Redis hot path instead of database reads in execution, queues/streams between components, C++ threading/atomics/mutexes where needed and batched durable writes outside the hot path.
- Modeled execution quality through preflight readiness, duplicate-intent checks, exchange gates, slippage constraints, fill/reject

journals, position state and replay analytics.

Strategy and validation surface

- Researched families across momentum, reversal, mean reversion, dip-buy/rise-sell behavior, volatility/options, breadth, regime detection, sector rotation, cross-asset relationships, portfolio/risk overlays and execution-aware trading.
- Built equity/futures relationship studies using market and sector anchors, side-specific rules, state-risk overlays and rolling train/test separation.
- Built options research pipelines for index/equity call-put workflows using OI, IV, Greeks-style features, expiry, moneyness, price movement and intraday option behavior.
- Built completed-prior-data walk-forward studies, fee-aware replay, paper-fill simulations, execution-variant experiments and reports for drawdown, win rate, profit factor, Sharpe/Sortino, fees, slippage, fill rate and capital usage.

Live/replay evidence

- Built live/replay infrastructure connecting archives, live-current signal files, model artifacts, strategy contracts, risk contracts, order intents, smart-execution journals, exchange journals and socket events.
- Generated structured JSON/JSONL lifecycle logs for signal generation, candidate selection, risk decisions, order intents, exchange submission paths, exit journals, smart-execution decisions and order-stream events.
- Tracked preflight readiness, duplicate-intent checks, exchange gates, slippage constraints, laddering fields, exchange-send status, side capital, projected exposure, rejection reasons and portfolio state.

ENGINEERING FOCUS

- Build systems with clear boundaries between research, simulation, replay, paper/live-runtime records and post-trade analytics.
- Prefer simple operational contracts: JSON messages, Redis hot state, PostgreSQL durable storage, replayable journals, heartbeats and observable runtime state.
- Optimize live paths by keeping active market state and working orders in memory, using queues between components and moving durable writes outside the execution path.

EDUCATION

B.Tech Information Technology, MPSTME.